Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Straight line <u>through the origin</u> [shows that acceleration is directly proportional to displacement] (1) Negative slope [shows that the acceleration is always directed towards the fixed point] (1)	2			2		
		(ii)		Calculating gradient i.e. 60 or obtaining suitable point from graph e.g. -0.05, 3.0 (1) Gradient = [-] ω^2 or substitution into equation (1) ω = 7.75 [rad s^{-1}] (1) **Allow 1 mark** for slips such as: $\omega = \sqrt{\frac{3}{0.5}}$ **Allow 2 marks** for $\omega = \sqrt{\frac{3}{0.5}}$ = 2.45 **or** $\omega = \sqrt{\frac{1.5}{0.03}}$ = 7.07 **Allow 3 marks** for Accept $\sqrt{\frac{1.75}{0.03}}$ = 7.64 [rad s^{-1}] **or** $\sqrt{\frac{2}{0.033}}$ = 7.8 [rad s^{-1}] **or** $\sqrt{\frac{2.4}{0.04}}$ = 7.85 [rad s^{-1}] **Allow 1 mark** for: $\omega = \sqrt{\frac{a_{\text{max}}}{A}}$ = 7.7 i.e. no evidence of data points used or substitution but method ok. **Alternative for full marks:** Using $a = \frac{v^2}{r}$ (which leads to v = 0.39) (1) Using $\omega = \frac{v}{r}$ (1) Convincing correct answer e.g. 7.75 or $\frac{0.39}{0.05}$ (1)		3		3	2	